
Mohammad Ali Safaei

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<https://github.com/mohsafaei>

EDUCATION

University of Tehran – Tehran, Iran (#322 in QS Rankings)

2021 – 2024

M.Sc. in Mechanical Engineering | Applied Mechanics specialization

Thesis Title: Distinctive Electromechanical Performance of Metamaterial Dielectric Elastomer Actuators with TPMS Geometries ([Link](#)).

Supervisor(s): Dr. Mostafa Baghani, Dr. Majid Baniassadi.

GPA: 18.15/20.00.

Bu-Ali Sina University – Hamedan, Iran

2016 - 2021

B.Sc. in Mechanical Engineering

Supervisor(s): Prof. Ali AlaviNia.

GPA: 15.66/20.00.

RESEARCH INTERESTS

- Linear/ Non-linear Finite Element Analysis
 - Constitutive Modelling, Multi-physics Modelling
 - Micromechanics, Multi-scale Modelling
 - Architected Materials Modelling
 - Machine Learning, Deep Learning,
 - Additive Manufacturing, 3D bioprinting
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SKILLS

Languages: Persian (Native), English: IELTS Academic: Overall: 7.5 |(L:7.0, R: 9.0, W: 7.0, S: 6.5).

Programming: Python, MATLAB, Fortran, JavaScript, MAPLE.

CAD software ([Link](#)): SOLIDWORKS ,CATIA.

CAE software ([Link](#)): ABAQUS, COMSOL Multiphysics, , LS-DYNA.

3D printing software: Simplify3D, Cura.

Documentation Software: MS Office (Word, PPT, Excel).

Hands-on Experiences: Casting, Welding, Machining, Turning, Milling, Nondestructive testing (NDT).

PUBLICATIONS

[M. A. Safaei, M. Baghani, M. Baniassadi, and M. Bodaghi, "In Silico Actuation Performance Investigation of Dielectric Elastomers with TPMS Geometries ",European Journal of Mechanics - A/Solids ,p. 105540, 2024.](#)

Multiaxial Finite Strain Behavior of Polydomain Liquid Crystal Elastomers (Proofing) –

DOI: <https://doi.org/10.21203/rs.3.rs-9517724/v1>

Book chapter: Toward Responsive Polymer Actuators Based on Novel Composite Structures – Publication:

August, 2026 | Link: <https://link.springer.com/book/9783032219763>

Review: Recent Advances in 3D/4D Printing of Soft Electroactive Polymers – (In Completion)

Data-Driven Surrogate Models Based on Convolutional Neural Networks Achieve High-Fidelity Prediction of the Stiffness Response of Hybrid TPMS lattices (In Completion)

EXPERIENCES

School of Mechanical Engineering, University of Tehran, Tehran, Iran

2022 – Now

| *Research Assistant - SMS lab* |

Supervised and mentored students in their projects and oversaw junior students in part-time **3D printing operations**.

Authored Section 1.3 of the book "**Computational Modeling of Intelligent Soft Matter**" ([Link](#)).

| *Teaching Assistant* |

FEM ([Link](#)), Continuum, Non-linear Solid Mechanics ([Link](#)) and Research Methodology ([Link](#)) Courses.

Delivered lectures and presentations, designed and assessed projects and exams, and provided feedback on students' "proposal writing" assignments.

Voluntary Job

2023 – Now

| Assistant Peer-Reviewer |

Assisted in the peer-review process for over 50 manuscripts in journals including *Materials & Design*, *Thin-Walled Structures*, *Chemical Engineering Journal*, *Additive Manufacturing*, *International Journal of Biological Macromolecules*, etc.

Emamat High School | Tehran, Iran

2024

| English Teacher (General) |

[Nogolearn](#) | Tehran, Iran

2025

| English Teacher (Upper-Intermediate /Advanced) |

IKCO Company | Tehran, Iran

2021

| Intern |

MDFICO – Tehran, Iran

2018 – 2020

| Junior Engineer/ Intern |

HONORS & AWARDS

Ranked 12th in the Nationwide PhD Entrance Examination (Solid Mechanics specialization).

Top 1%, out of more than 10,000 applicants in the M.Sc. Nation-wide University Entrance Exam (Full scholarship).

Top 5%, out of more than 190,000 applicants in the B.Sc. Nation-wide University Entrance Exam (Full scholarship).

Awarded within the top state candidates in the National Kharazmi Elite Youth Festival. (Poetry-Literature)

Performed as a Santur player in the talented students orchestra during elementary and secondary school.

LICENCES & CERTIFICATES

ICDL - Python: Fundamental & Advanced - COMSOL: Multiphysics - MATLAB: fundamental - Piping principles and design - Pneumatic Technician - ISO 9001:2015 Internal Auditor – SEM & TEM

ACADEMIC PROJECTS

Artificial Intelligence / Neural Networks ([Link](#))

- Implemented machine learning algorithms (KNN, SVM, Logistic Regression) for pattern recognition on benchmark datasets.
- Developed image classification models using CNNs and MLPs on a traffic sign dataset.
- Applied reinforcement learning algorithms (Monte Carlo methods, Deep Q-Networks) to solve the Taxi environment problem.

Viscoelasticity ([Link](#))

- Implemented Quasi-Linear Viscoelastic (QLV) models to simulate benchmark problems including stress relaxation & creep response.
- Investigated the influence of glass transition temperature on rubbery plateau behavior using the WLF equation.
- Modeled the visco-hyperelastic behavior of hybrid TPMS lattice structures under thermomechanical loading conditions.

Nonlinear Solid Mechanics Course ([Link](#))

- Developed MATLAB code for simultaneous calibration of constitutive models using multiaxial experimental data.
- Formulated semi-analytical constitutive models for transient and quasi-static swelling of pH- and temperature-responsive hydrogels.
- Built numerical models for shape memory polymers (SMPs) with lattice architectures under thermomechanical loading.

FEM: Intro & Advanced Courses ([Link](#))

- Developed UEL, UMAT, and UHYPER subroutines in ABAQUS for various hyperelastic constitutive models from the literature.
- Implemented the finite element method from scratch in MATLAB and Python for linear elasticity problems.

Biolographic review ([Link](#))

- Conducted a systematic bibliographic review and literature analysis on smart polymer composite actuators.

References

- Mostafa Baghani- Professor – University of Tehran (Iran) - [email](#)
- Majid Baniassadi- Professor – University of Tehran (Iran) - [email](#)
- Mahdi Bodaghi – Associate Professor - Nottingham Trent University (UK) - [email](#)